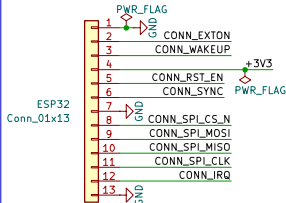
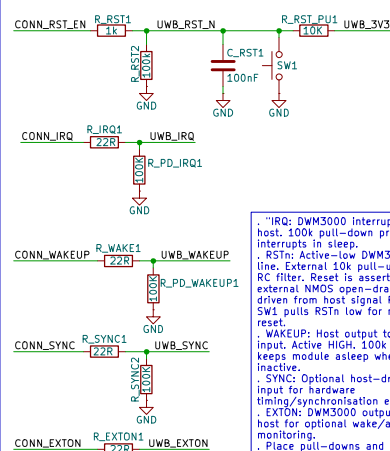


INTERFACE CONNECTOR



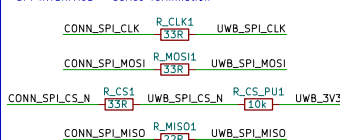
• "Host connector for ESP32 or other MCU. Pinout: 1:GND 2:EXTON 3:WAKEUP 4:RST_EN 5:SYNC 6:+3V3 7:GND 8:CS_N 9:MOSI 10:MISO 11:CLK 12:IRQ 13:GND
• EXTON is a DWM3000 output for wake/activity monitoring.
• RST_EN drives the external NMOS reset stage (HIGH at host = reset active on DWM3000).
• Keep SPI wiring short and route with nearby GND where possible."

CONTROL SIGNALS – IRQ, RSTn, WAKEUP, EXTON



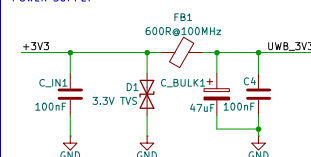
• "IRQ: DWM3000 Interrupt output to host. 100k pull-down prevents false interrupts in sleep.
• RSTn: Active-low DWM3000 reset line. External 10k pull-up + 100nF RC filter. Reset is asserted by an external NMOS open-drain stage driven from host signal RST_EN. SW1 pulls RSTn low for manual reset.
• WAKEUP: Host output to DWM3000 input. Active HIGH. 100k pull-down keeps module asleep when host is inactive.
• SYNC: Optional host-driven sync input for hardware timing/synchronisation experiments.
• EXTON: DWM3000 output routed to host for optional wake/activity monitoring.
• Place pull-downs and EXTON/SYNC series resistors close to the DWM3000 side."

SPI INTERFACE – Series Termination



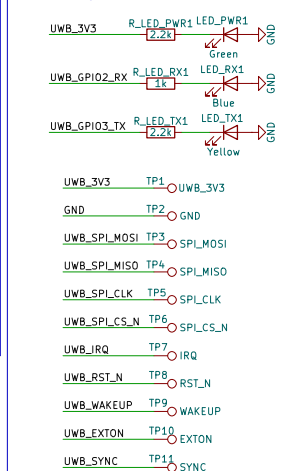
• "R_CLK1, R_MOSI1, R_CS1: 33k series resistors on ESP32-driven SPI lines. These reduce ringing and help signal integrity over connector/cable interconnects.
• R_MISO1: 22k series resistor on DWM3000-driven MISO line. Improves signal integrity and reduces reflections during bring-up and normal operation.
• R_CS_PU1: 10k pull-up holds SPICSn HIGH while the ESP32 is reset, disconnected, or booting.
• SPI mode is fixed in hardware: SPIPHA = 0, SPIPOL = 0; => SPI Mode 0 (CPOL=0, CPHA=0)
• Use 1–2 MHz SPI during initial bring-up, then increase speed after stable DEV_ID reads.
• Place R_CLK1, R_MOSI1, and R_CS1 close to the host/connector side.
• Place R_MISO1 and R_CS_PU1 close to the DWM3000 side."

POWER SUPPLY



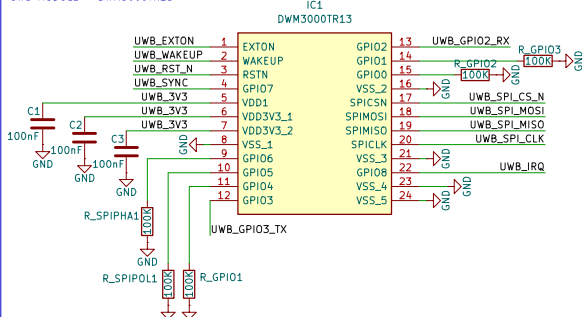
• "FB1: 600Ω @ 100MHz ferrite bead isolates DWM3000 switching/RF noise from the main 3V3 rail. If FB1 is unavailable during bring-up, it may be temporarily replaced by a 0Ω link or wire bridge.
• D1: 3.3V TVS diode on the input side for ESD and transient protection.
• C_IN1: 100nF input decoupling before FB1.
• C_BULK1 + C4: 10uF + 1uF local bulk decoupling on UWB_3V3 after FB1.
• Additional 100nF capacitors are placed close to the DWM3000 VDD pins in the module block. Place local decoupling capacitors close to the module and keep GND return paths short."

DEBUG – Diagnostic LEDs & Test Points



• "Diagnostic LEDs are fitted by default for bring-up and communication tests.
• R_LED_PWR1: 2.2kΩ, green LED, indicates UWB_3V3 present.
• R_LED_TX1: 2.2kΩ, yellow LED, driven from DWM3000 GPIO3 when configured as TXLED in firmware.
• R_LED_RX1: 1kΩ, blue LED, driven from DWM3000 GPIO2 when configured as RXLED in firmware.
• LED anodes connect to the signal through the resistor; cathodes connect to GND. Mark LED cathodes clearly on silkscreen to avoid reversed assembly.
• Test points: TP1 = UWB_3V3, TP2 = GND, TP3 = SPL_MOSI, TP4 = SPL_MISO, TP5 = SPL_CLK, TP6 = SPL_CS_N, TP7 = IRQ, TP8 = RST_N, TP9 = WAKEUP, TP10 = EXTON, TP11 = SYNC.
• Place test points on an accessible PCB edge for oscilloscope and logic analyzer probing."

UWB MODULE – DWM3000TR13



• "DWM3000TR13 core module connections.
• Power: VDD1, VDD3V3_1, and VDD3V3_2 are powered from UWB_3V3. C1, C2, and C3 are 100nF local decoupling capacitors. Place each decoupling capacitor as close as possible to its respective VDD pin.
• SPI: SPICSn, SPIMOSI, SPIMISO, and SPICLK connect to the host SPI interface.
• IRQ: GPIO6/IRQ routed to host interrupt input.
• GPIO straps: GPIO6 / SPIPHA pulled LOW with 100kΩ => SPIPHA = 0, GPIO5 / SPIPOL pulled LOW with 100kΩ => SPIPOL = 0; => SPI Mode 0 selected at power-up.
• Optional GPIOs: GPIO2 and GPIO3 are used for RX/TX diagnostic LEDs. GPIO7 is routed as UWB_SYNC for optional hardware synchronization and debug.
• Reminder: Observe DWM3000 antenna keep-out rules in PCB layout. Do not place copper, ground, vias, battery, or metal objects in the antenna keep-out area."

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